Evaluating Tropical Cyclone Tracks with TempestExtremes

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Research Proposal

Abstract

I will evaluate TempestExtremes (TE) tracks for Ida, Larry, and Nicholas in 2021 using ERA5 and IBTrACS. Python scripts will prepare inputs, run three pressure contour thresholds, associate tracks, and summarize recovery and position separation. An existing Ida trial demonstrates feasibility. The project will examine how parameter choices affect recovered track segments and produce a reproducible case study with documented settings and limits on interpretation.

Background and motivation

Tropical cyclone (TC) statistics from reanalyses and climate models rely on automated trackers. TE extracts storm trajectories using configurable physical and linking criteria [1, 2]. Earlier work shows that these choices matter. Detection thresholds need to suit the data resolution [3], and changing TE's criteria changes which TCs and track segments are identified [4]. Different tracking schemes can disagree on the same simulations, and threshold selection is key to reproducible results [5]. In ERA5, four trackers each detected about 80% of observed storms, but their TC frequencies and durations differed [6]. Reanalyses contain nearly all observed TCs but underestimate their intensity, and identification is least certain for weak storms [7]. A pressure threshold may therefore matter most at weak or early stages. I want to understand this sensitivity before using detected tracks in later research. Three selected cases allow individual missed positions to be inspected within a manageable programming project.

Research questions and hypothesis

R1. How closely do baseline TE tracks follow IBTrACS, and which reference positions are recovered? **R2.** How do 1-, 2-, and 3-hPa pressure contour thresholds change recovery and separation when other settings remain fixed?

I hypothesize that a weaker threshold may recover more weak or early-stage positions but also introduce competing tracks. A stronger threshold may omit those positions. Both recovery and separation are needed to assess the result.

Data and study area

Downloaded ERA5 fields cover 24 August–17 September 2021 at six-hourly UTC times on a 0.25° grid [8]. Inputs include sea-level pressure, 10-m winds, geopotential at 300 and 500 hPa, and surface geopotential. IBTrACS v04r01 supplies reference positions [9, 10]. Figure 1 shows the selected tracks within the ERA5 domain, 0–70°N and 110°W–10°E. Available storm stages are retained and will inform interpretation of missed positions.

Methods and programming plan

Prepare consistent inputs. I will extend the existing Ida Python workflow to the complete downloaded period. With xarray and Pandas, I will standardize coordinate names, UTC timestamps, longitude conventions, and units; separate the two pressure levels; and convert surface geopotential to height. Checks will identify duplicate or missing timestamps, nonfinite fields, and incorrect dimensions before TE is run. Processing the regional subset keeps memory and runtime manageable.

Run controlled configurations. The installed pilot uses TE 2.4.2 and the tropical cyclone recipe described by Ullrich et al. [2]. I will write a parameterized Python runner for DetectNodes and StitchNodes, changing only the sea-level pressure contour increment: 100, 200, or 300 Pa, equivalent to 1, 2, or 3 hPa. The contour radius will stay at 5.5°. Other baseline criteria, including the warm-core test, wind and terrain filters, 8° linking range, 54-hour minimum track duration, and 24-hour maximum gap, will remain fixed. The runner will create a separate output directory for each setting and save commands, software version, exit status, and logs. The three configurations will each be evaluated

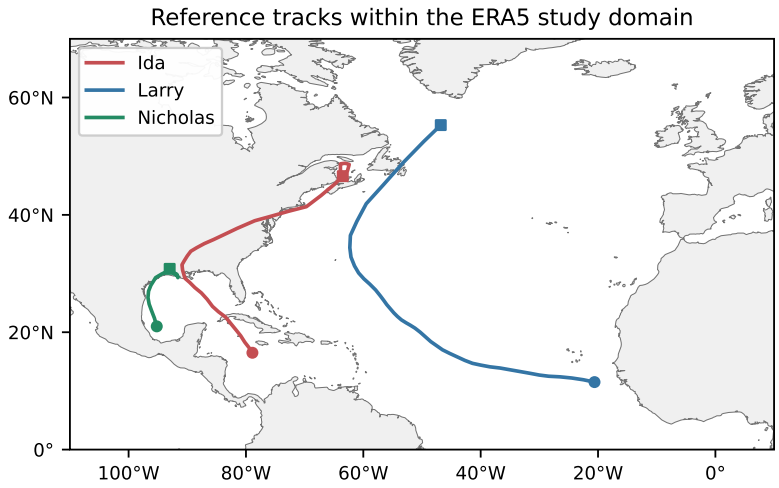


Figure 1: IBTrACS tracks for Ida, Larry, and Nicholas at six-hourly times within the downloaded ERA5 period. The map spans the ERA5 domain. Circles mark the first retained positions and squares the last. Tracks were plotted from the saved IBTrACS subset [10].

against all three storms.

Associate tracks with reference storms. Before inspecting the sensitivity results, I will save a fixed reference table for each storm using valid six-hourly IBTrACS positions within the downloaded time and spatial domain. For Ida, I will retain the existing 26 August 12:00–1 September 18:00 UTC window so the pilot comparison remains reproducible. There will be no time interpolation. For each TE track, I will join positions to the reference by timestamp and calculate spherical great-circle separation using a mean Earth radius of 6371.0088 km. Spherical distance is consistent with TE's geographic calculations [2].

A pair will count as matched when its separation is at most 300 km. A candidate track must contain at least three matched times. I will select the candidate with the most matches, using lower median matched separation to break ties, and flag competing candidates for visual review. If no candidate qualifies, recovery will be zero and position summaries will be undefined. The 300-km gate is an association choice for this project, not a demonstrated accuracy standard, and will remain fixed across pressure settings.

Answer the research questions. For R1, I define recovery as $C = N_{\text{matched}}/N_{\text{reference}}$, where the denominator includes all valid positions in the predetermined reference window. I will report both counts with the percentage, plus the mean and median separation at matched times. Missing times will remain explicit in the comparison table. These operational summaries complement the configurable detection criteria [2, 4]; matched distances alone exclude missed positions and cannot describe overall recovery.

For R2, Python will combine the nine storm–configuration comparisons into one table and generate maps, separation time series, and plots comparing settings. I will inspect where coverage changes and whether the difference comes from weak stages, linking gaps, or a changed track association. These are descriptive case comparisons; three storms cannot support basin-wide performance claims. Other detected regional tracks will not automatically be counted as false alarms because they have not all been evaluated against a complete reference inventory.

Verify and preserve the analysis. I will check the distance function with coincident points and a known angular separation, confirm one record per timestamp, and manually inspect selected matched and missed positions. Scripts, configuration files, reference windows, figures, and summary tables will be retained. AI assistance may support coding and debugging, with prompts and changes recorded; numerical claims will be checked against the saved inputs and outputs.

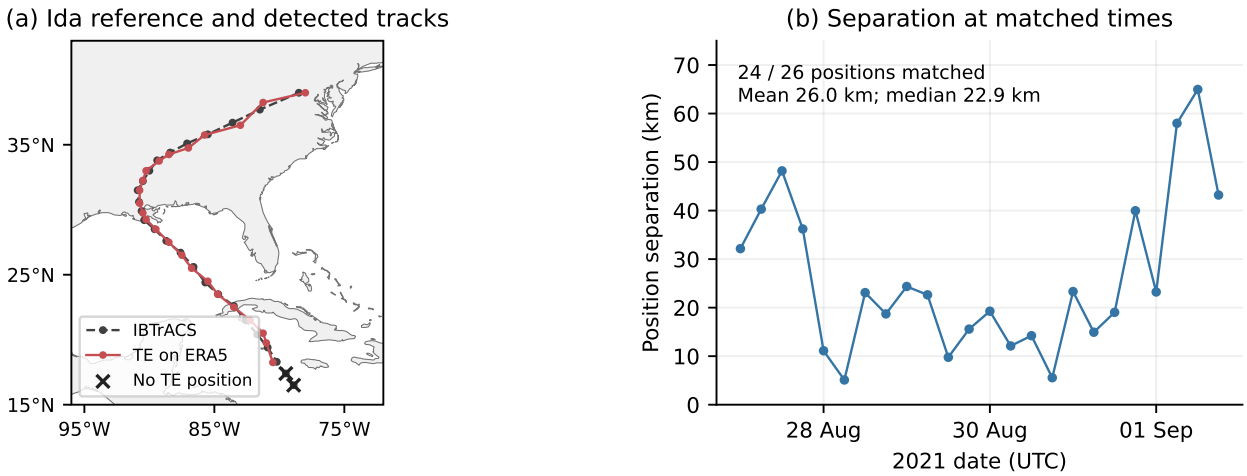


Figure 2: Original preliminary result for Ida, 26 August 12:00–1 September 18:00 UTC, using the 2-hPa baseline. (a) IBTrACS and TE paths; crosses identify the two reference positions without a TE position. (b) Center separation at matched times. The two missing times do not contribute to the distance averages. These results describe this window only.

Expected contribution and broader impacts

The outcome will be a reproducible example showing how a pressure criterion changes recovered storm segments and how to evaluate that change in Python. Transparent settings and explicit missing positions can help later users judge whether a tracker is suitable for climate or hazard research. The results will be limited by the selected storms, the regional window, and ERA5 resolution. I will not interpret ERA5 gridded wind maxima as equivalent to IBTrACS sustained-wind estimates or claim operational forecasting skill.

Preliminary result and feasibility

The saved Ida baseline run recovered 24 of 26 reference positions (92.3%) using the 300-km gate. Mean matched separation was 26.0 km and the median was 22.9 km. I recalculated these statistics from the saved position pairs for this proposal. The two missing positions occur at the start of the window (Figure 2). This establishes that input preparation, TE execution, and comparison work for one case. The other storms and threshold experiments remain proposed work.

Timeline and deliverables

During 19–25 October, I will finalize the proposal and save the reference windows and figure inputs. From 26 October–8 November, I will extend data preparation and the baseline comparison to Larry and Nicholas. During 9–22 November, I will implement the configuration runner and complete the three pressure settings. From 23 November–6 December, I will inspect changed track segments, generate comparison figures, and write the report. I will reserve 7–10 December for rerunning checks and final revision. The deliverables will be the Python workflow, saved settings and logs, a comparison table, figures, and a report. If time is tight, I will complete these pressure experiments before considering any additional parameters.

References

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